

Inventory system of Ranz photography stocks



**A System Analysis and Design Presented to the
Faculty of the Institute of Computing
Davao del Norte State College
New Visayas, Panabo City**

**Cadavos, Christine Joy
Madronero, Christine Joy M.
Sardual, Rheza D.
Verano, Marilou A.**

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ABSTRACT

This research study focuses on eliminating the hindrance of Ranz photography in their manual inventory of stocks. The organization poses great growth in adapting through the years passed, the owner stated to eliminate such problems in the organization which the researchers proposed to do. By the implemented system, the owner could oversee the list of items needed to be purchased, including the total summary of the stocks or items left are also can be seen in the system.

Moreover, the possibilities of eradicating misinformation which could lead to improper reports and purchased wrong items, this also removes the hassle to the owner of summing up all of the stocks through writing down the whole inventory. Through inputting the data's needed to build this system, it also has benefits for the knowledge of individuals to serve as guide for future outcomes. Furthermore, this research could help the organizations to have an overview of the future preference in improving and adapting to the new normal of using systems as a substantial growth.

ACKNOWLEDGEMENT

The researchers would like to thank and have this opportunity to express their heartfelt gratitude to everyone who helped them in completing this study. First and foremost, researchers would like extend their deepest thanks to their parents and family members for their unwavering support, encouragement, and financial assistance were this is crucial in keeping us motivated and focused throughout this journey. Second, they would like to acknowledge to their project instructor which we have been greatly guided throughout the entire process. Despite her busy schedule, she always found time to help, and her dedication significantly contributed to this research study. They would also like to thank the Ranz photography for the participation in our research study, your cooperation, effort, and willingness to engage made a significant difference in the success of our study.

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CHAPTER I

INTRODUCTION

A. Company Overview

Ranz Photography that is located at Quezon St. New Pandan, Panabo City. The owner stated that this organization was established in January 9, 2017, which has been reaching high to be a successful business. This business offers services of taking photos, tarpaulins, photo frames, customized mugs, customized Stickers and ID laminations, they accept walk in costumers, and also they have a Facebook page which also entertains costumer through online service. Moreover, this organization is open in Monday to Saturday in 9am to 5:30pm while in Sunday is 9pm to 3pm. They also had 9 employees in total including the owner and his assistant. The owner stated that they a lot. This organization has been also adapting to trends that helps people in photo process and including taking pictures for memories.

B. Organizational Structure

The Ranz Photography has its owner who leads the organization, one employer accounted as its assistant that will oversee the works of other employees, and seven employees who will perform the services an organization offered. An organization has its total number of nine people who works and offered for services.

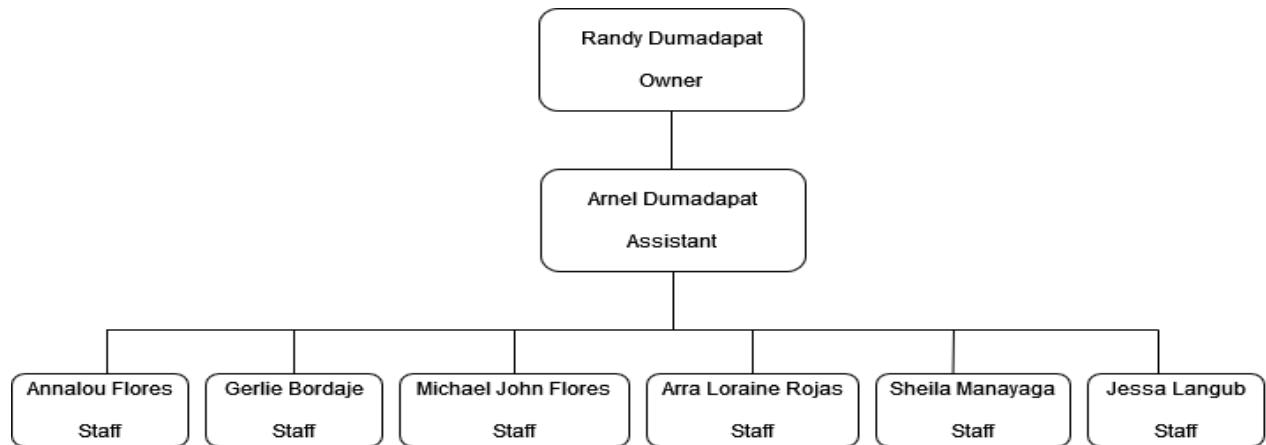


Figure 1. Organizational Structure

C. Problems Encountered

- Manually writing down the remaining stocks- This is time consuming and the possibilities that could cause misinformation.
- Knowing the stocks they need to purchase- They don't know the specific stocks to purchase that could affect their stocks inventory.
- Monitoring the orders and supplies for processing services- They could prepare the budget for purchasing the stocks in the future.

D. Existing Business Process

Ranz photography accepts the customers demand through their staffs. The process of accepting the demand services, starts in processes the specific service the customer demands and through the computer of staffs they edited the specific image or word the customer demands to be printed. Alongside they will let the customer know the possible changes they want and proceed to process the service in the corresponding printing machine. The transactions will be done after the customer pays for the service provided by the organization.

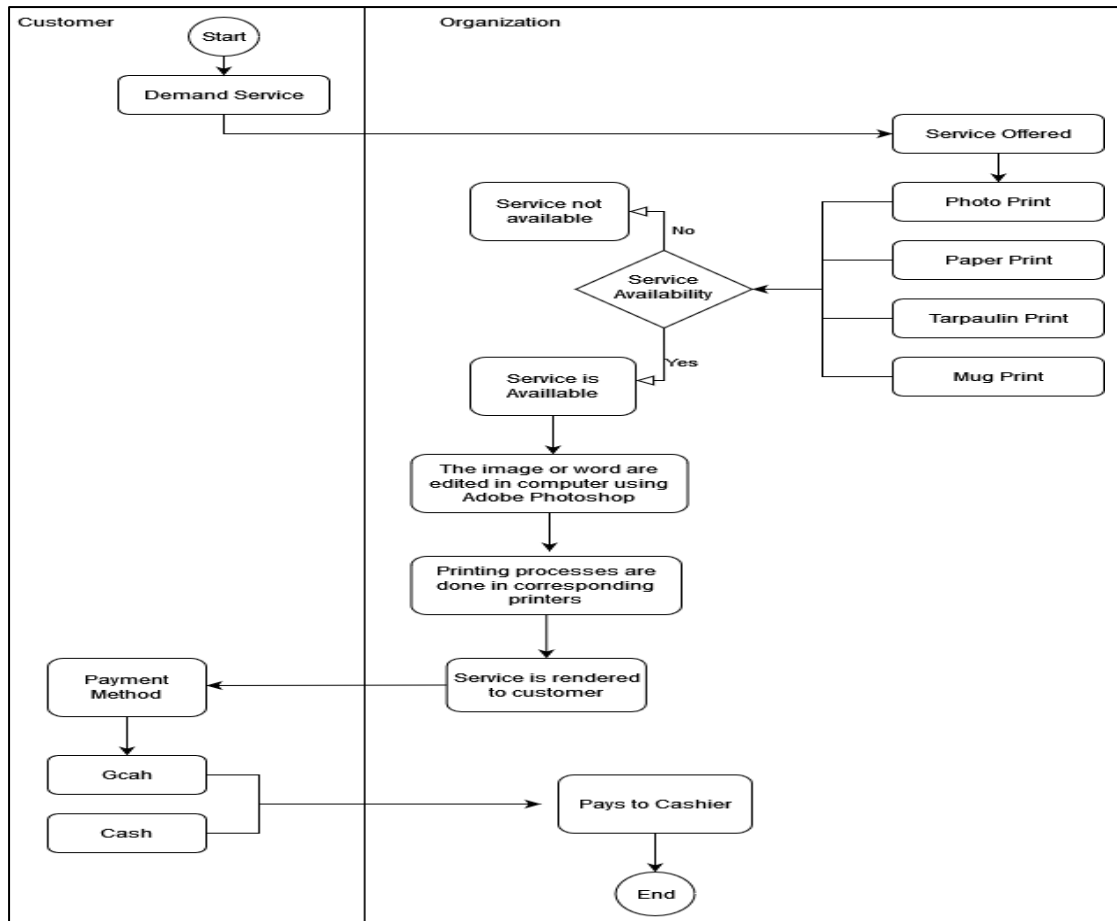


Figure 2. Existing Business Process

RANZ photography also offers products such as mugs and photo frames. The organization usually offers these products when the printing services needed a photo frames and mugs especially in the customization like mug printings and photo shoots of customers.

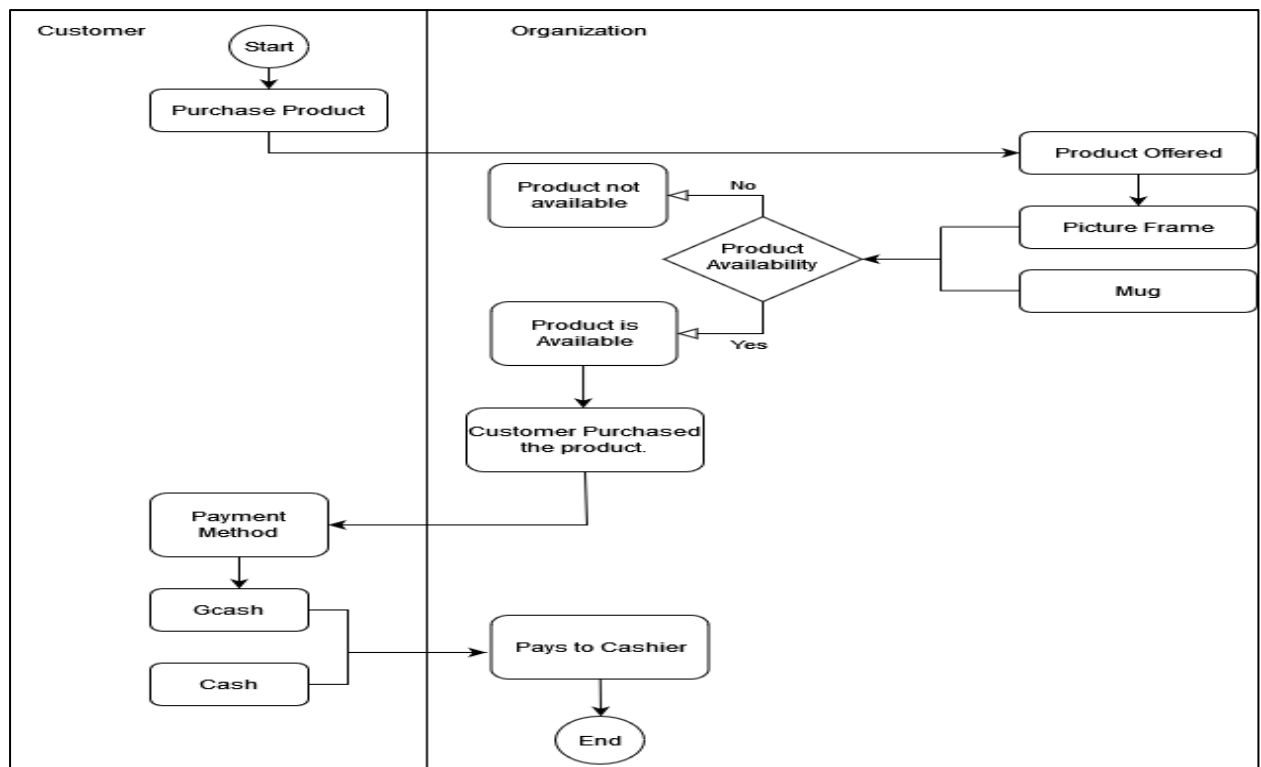


Figure 3. Existing Business Process

Upon entering the organization they will see an available services in the bulletin board including the items available in the counter like picture frame and mug which they will also see if it is available or not. This follows in how they purchase the product and next is payment method which they also accept Gcash payment. They will receive the service and items after they pay.

E. Objectives of The Project

General Objectives

The objectives is to develop a system for Ranz Photography that can eliminate the problems of manually recording the inventory which will improve the records of stocks and their business process

Specific Objectives

This section expound on the specific aims of the project. It also presents the project's detailed objectives.

1. Input product data of inventory stocks.
2. System could compute the remaining inventory stocks.
3. Display the entirety flow of inventory
4. Generate the whole inventory reports of stocks.

F. Scope and Limitations

This study will develop the Ranz Photography's inventory system which will eliminate their manual process of records in stocks. The people has access in the system which will help them in getting the records of stocks and products. Hardware helps the people in accessing the system including the Software in creating and developing the system. The Data follows on what the system offers and the Process is what could the system does

Table 1. Scope and Limitations of the Project

People	• Owner- Reviewing the inventory of stocks • Assistant/Staff- Creating the sum up reports
Technology	Hardware: • Wireless Fidelity - Help access the internet • Laptop- Access the system, Help to monitor the stocks Software: • Hyper Text Processor- programming language used in creating codes needed for building website development of the system. • XAMPP- this will help to test the codes • Sublime – this is where the system is build using the programming language
Data	• Inventory Data- Records of stocks and its prices including the products availability. • Product Data- Records of all products and stocks that is needed to be sell and purchased.
Process	• Process of products- To know the records of products and stocks that are available. • Inventory management process- Process of tracking down the stocks that is needed and purchased. • Process of generating inventory- This provides the entirety reports of flow in inventory

G. Related Systems and Works

The international study of the inventory system in food flavor manufacturing industry is an industry that looks for cost saving methods and is continuously looking at improving the quality of their products and satisfying their customers. The industry faces obsolescence of raw materials due to, wrong inventory, not managing the inventory levels as efficiently as they should as well as competitors introducing same quality materials at low cost prices. According to Evans superior quality, consistent

service, and competitive pricing are just an entry to the sales game. Porter, states that competitors usually play the strategy game according to different rules, making it harder to respond effectively. Therefore in order to compete, it is important to improve operational methods to ensure that they are effective to support implemented quality and cost reduction initiatives thus ensuring more productivity. Continuous improvement is an ongoing effort to improve products, services or processes. These efforts may seek "incremental" or "successful" improvement over time, as suggested by Adams [1].

The national Inventory of Manila is carrying cost in what is linked with the amount of inventory either raw materials or finished goods products stored in the warehouse. Capital expenses are the largest contributor in our inventory carrying cost because it includes the purchased prices of the materials that we are keeping in the warehouse. This is the cost of the volume that directly affects the price and this is the available inventory that is currently held in stock. The company refers to this volume to project how much profit can be materialized on the current inventory level and how much more the company plan produced or manufactured Inventory service costs are the expenses from the insurance, systems used to process the order or to monitor the inventory, and to the employees who perform periodic inventory count, etc. This cost is the sum of expenses incurred in placing and processing the orders. Inventory risk costs are the costs that result from losses due to theft, wear and tear, and depreciation. A stock-out is a situation where internal or external demand cannot be met from current inventory. Costs that contribute to internal costs are (process delays, wasted work, unplanned or lost production, etc.). All these costs can be eliminated through effective inventory management control and good inventory turnover. Undoubtedly, the proper planning

of safety stock Replenishment cycle is one of the most important factors of effective stock recording. There are various objectives based on which an optimized safety stock is created, such as minimizing costs, maximizing service levels and general considerations. For practical applications, optimal identification approaches based on cost and service level objectives are more appropriate [2].

A local study of inventory system in Mindanao is a useful tool for keeping track of and monitoring various products that are brought into and taken out of a business's warehouse or location, mainly for bookkeeping needs. Maintaining an account of all stocks requires a corporation to keep an eye on all transactions and the flow of commodities. Nevertheless, in certain organizations, the inventory of items is done manually, meaning that each time a material is brought into or taken out of a room, an employee notes its details. The similar situation occurs in Mindanao Kokusai Daigaku, which in the inventory system becomes more labor-intensive and prone to human error, which would probably result in a loss [3].

CHAPTER II

METHODOLOGY

A. Project Development Methodology

The project team choose the modified waterfall for the system methodology. The modified waterfall model provides an orderly sequence of development steps with some flexible, iterative stages to facilitate change called a “feedback system” between stages. So if any problem is found at one phase, the feedback system helps previous phase to correct an error or problem the system might have possibly encountered through the process. The modified Waterfall model consists of the same phases as the classic model but allows for phases to overlap and break the project down into subprojects.

The project team choose the modified waterfall for the system methodology because this procedure impacts both the order and procurement processes. The Modified Waterfall Approach of System Development Life Cycle (SDLC) was selected because it is appropriate for processes that are well-understood and allow overlapping phases. Moreover, provides the orderly sequence of development steps with some flexible, iterative stages to facilitate change called a "feedback system" which helps the previous phase to correct an error or problem of the system might have possible encountered through the process.

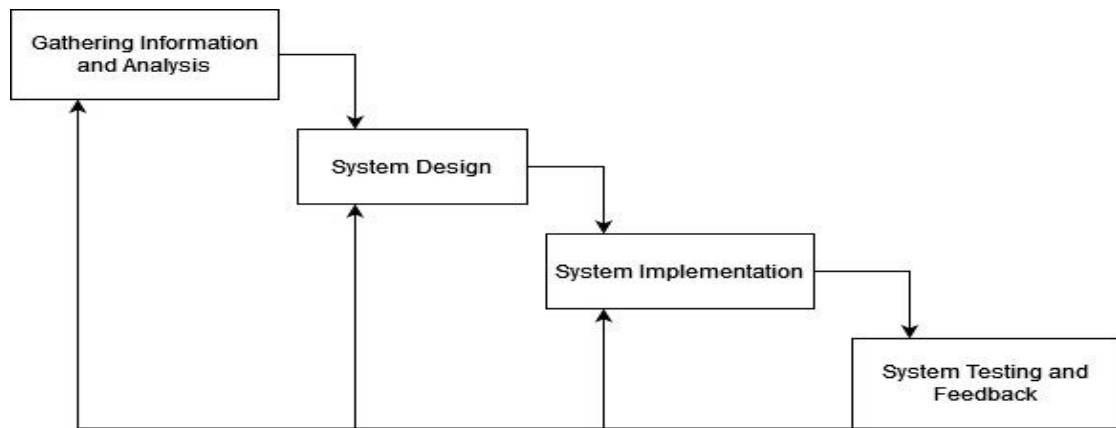


Figure 4. Modified Waterfall

Work Breakdown Structure

The project team composed a work breakdown structure for Ranz Photography Stocks Inventory System composing of four respective phases. The Gathering Information and Analysis phase consist of four respective activities starting in project team creating project plan for the system, preparing for the budget and the future needs of the system. Project team choosing and meeting the project sponsor and gathers data that the system needed.

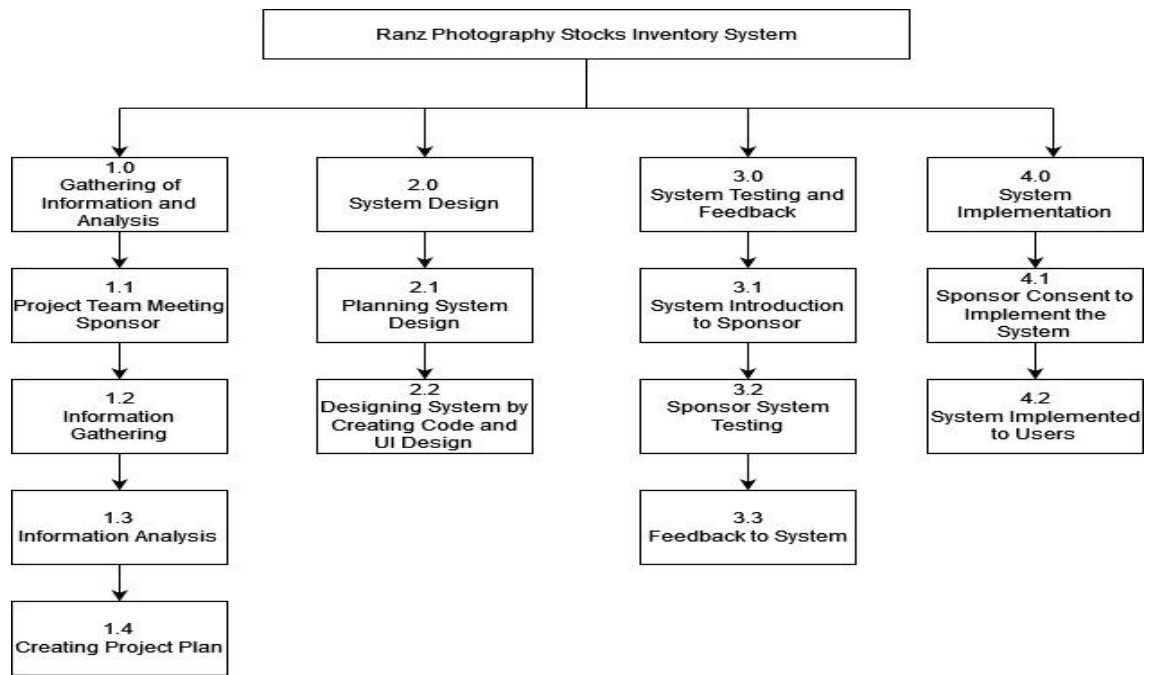


Figure 5. Work Breakdown Structure of the Project

Gantt Chart

Gantt Chart in figure below shows the timeline upon developing the system, it also indicate the specific dates in the corresponding phases of the system.

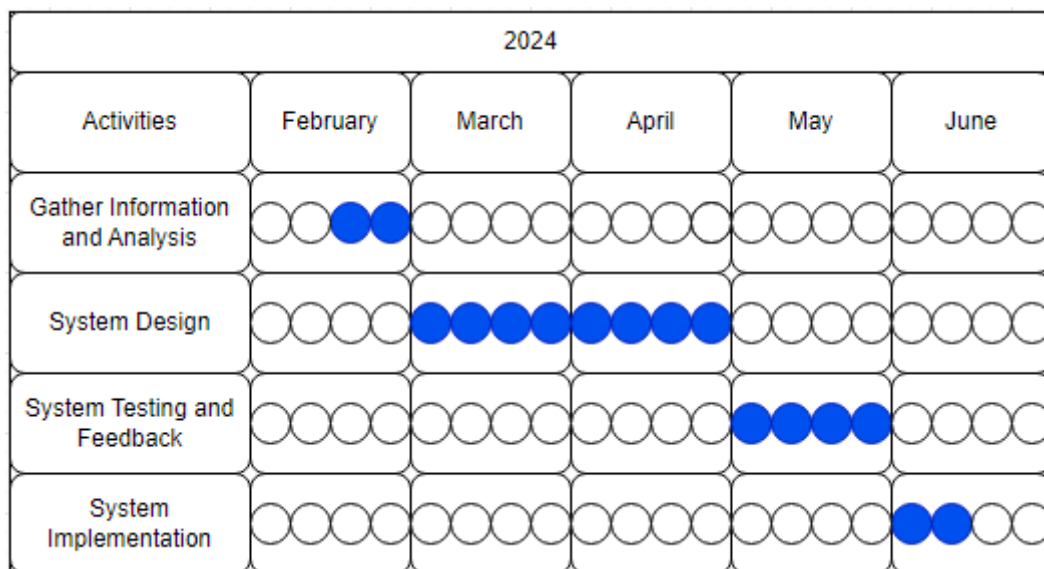


Figure 6. Gantt Chart of the Project

B. System Analysis

The system will provide a service that Ranz photography needed in improving their inventory of stocks that will eliminate the manual writing of stocks in every month. This will serve as their stepping stone in the photography industry which will help them to manage their stocks.

System Requirements

System requirements are detailed statements of the system's actions to satisfy stakeholder needs and requirements. These are derived from business and user requirements and are categorized into two: functional and non-functional requirements. These should be reflected in bullet form. It will include **functional** and **non-functional requirements**.

Functional Requirements:

- The system can oversee the inventory stocks
- The system can overview the total number of items left
- The system can generate inventory of stocks report
- The system can add and remove items.

Non-Functional Requirements:

Technical:

- The system doesn't have usability guidelines.

Performance:

- The system could not tell the prices and its total of the items.

Usability:

- The system cannot guide you upon entering the dashboard.

Security:

- The system should have tight privacy in signing in.

Process Model: Use Case Diagram and Description

The owner can access in updating the list generate the report and including overseeing the list but only the assistant will categorized the list. Moreover, both owner and assistant can also remove stocks also in adding stocks.

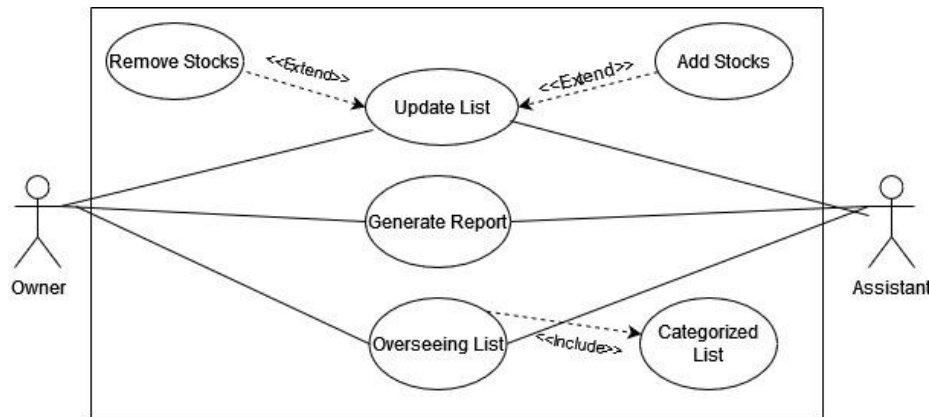


Figure 7. Use Case Diagram

The table below stated the usage of every use case in the Figure 7

Table 2. Use Case Description for Update List

Use Case Name	Update list
Participating Actors	Owner Assistant
Entry Conditions	From adding stocks and removing stocks they can update the list
Flow of Events	1. After seeing the updated list they can generate the report and they can oversee the updated report list
Exit Conditions	Categorized list
Quality Requirements	Efficient

Table 3. Use Case Description for Generate Report

Use Case Name	Generate Report
Participating Actors	Owner Assistant
Entry Conditions	• Provide overall inventory reports of stocks
Flow of Events	1. It generate the report and they can oversee the updated report list
Exit Conditions	• Generate the updated Stocks List
Quality Requirements	• Accuracy

Table 4. Use Case Description for Overseeing List

Use Case Name	Overseeing List
Participating Actors	Owner Assistant
Entry Conditions	• Provide overall details of the stock inventory
Flow of Events	1. After generating updated details of stocks inventory, this will oversee the updated report
Exit Conditions	• Performing other actions in the system
Quality Requirements	• Efficiency and effectiveness

Table 5. Use Case Description for Remove Stocks

Use Case Name	Remove Stocks
Participating Actors	Owner Assistant
Entry Conditions	Removing stocks from list of inventory
Flow of Events	1. It will remove the stocks in the inventory report of the system
Exit Conditions	After removing stocks, it will updated in update list
Quality Requirements	Efficiency

Table 6. Use Case Description for Add Stocks

Use Case Name	Add Stocks
Participating Actors	Owner Assistant
Entry Conditions	Adding stocks from list of inventory
Flow of Events	1. It will add the stocks in the inventory report of the system
Exit Conditions	After add stocks, it will updated in update list
Quality Requirements	Efficiency

Table 7. Use Case Description for Categorized List of Stocks

Use Case Name	Categorized List Of Stocks
Participating Actors	Owner Assistant
Entry Conditions	Classified stocks details
Flow of Events	1. Collects stocks details and classifies according to its category
Exit Conditions	Performing other actions
Quality Requirements	Efficiency and accuracy

Process Model: Activity Diagram

Logging in the details with username and password to enter the system. After logging in they can oversee the stocks remaining including the availability, they can add and remove stocks whenever they like. Furthermore they can have the updated list after they remove and add the stocks, also they can generate the report of stocks.

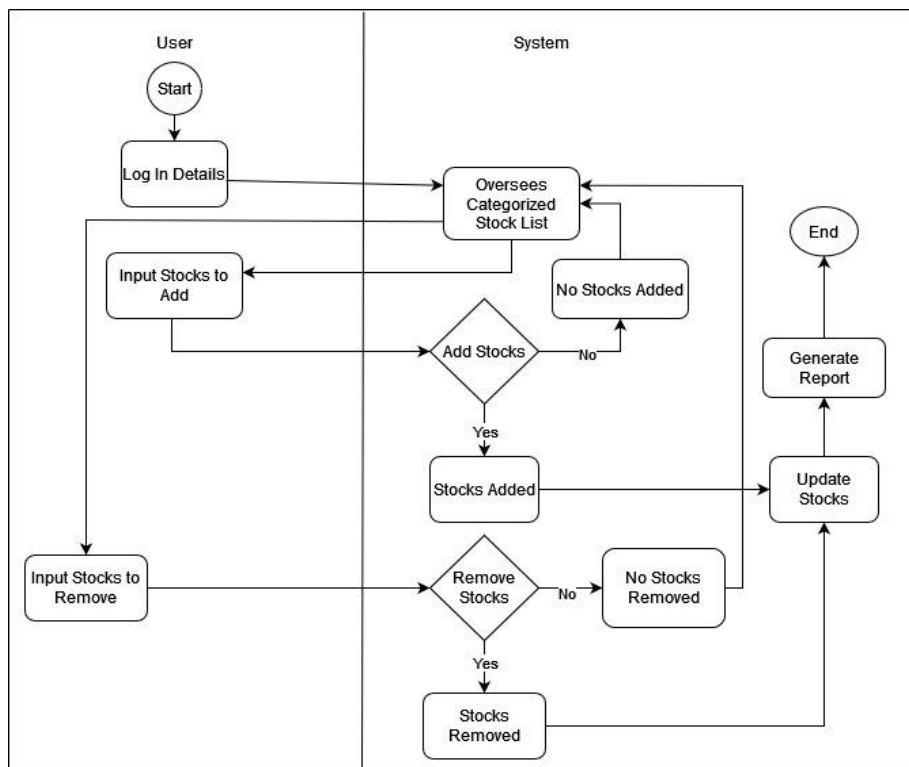


Figure 8. Activity Diagram

Data Model: Context Data Flow Diagram

The users can log in using the username and password, this follows in adding the stocks and removing it. After they enter the inventory system the users can access the stocks added and removed including the overview of the stocks, updated stocks and also generating the report.

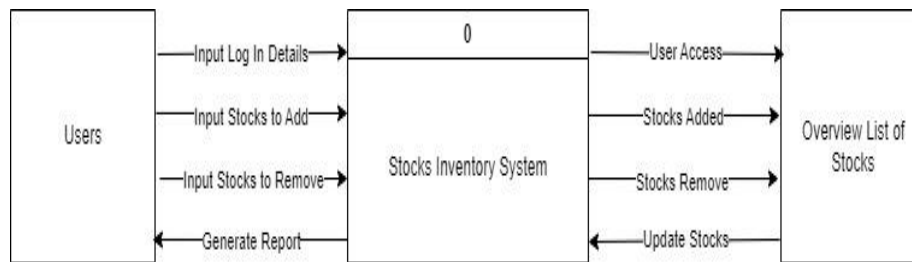


Figure 9. Context Data Flow Diagram

Data Model: Data Flow Diagram

The logged in the system to have access with provided details and overview the list of stocks which is categorized and its availability. Inputting the data to add stocks, including the removing of stocks which could lead to updating the stocks to generate report.

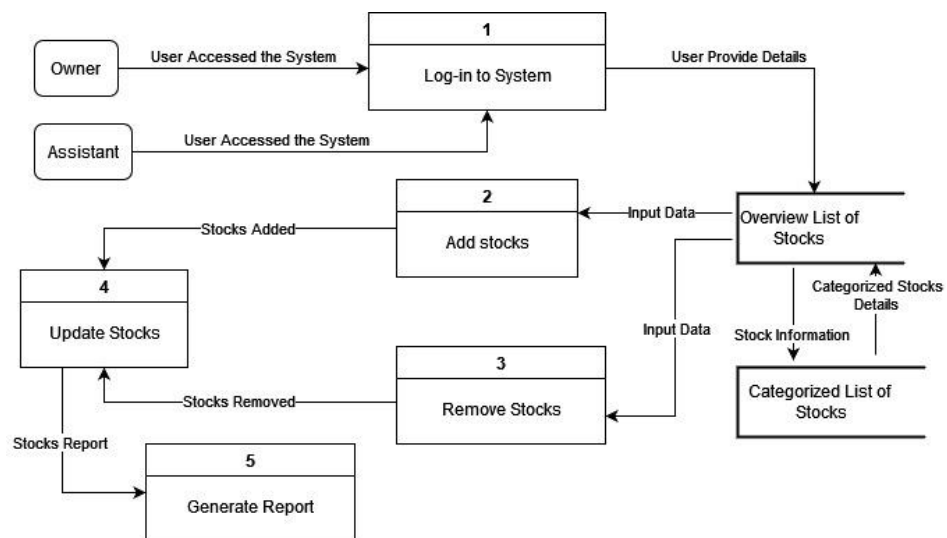


Figure 10. Level 1 Data Flow Diagram

Database Design: Entity-Relationship Diagram

This section presents the proposed ERD of the information system. Use the tool used for designing ERDs.

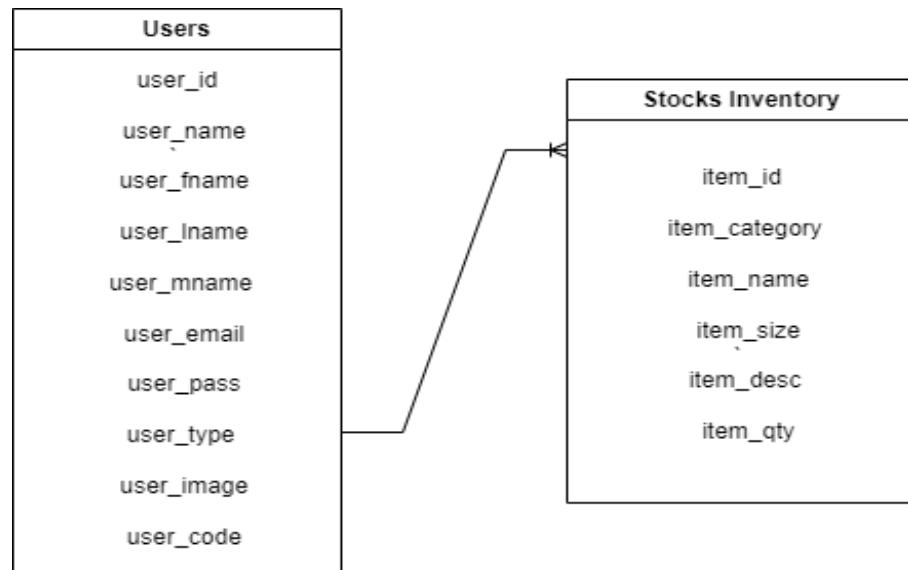


Figure 11. Entity-Relationship Diagram

Data Dictionary

In this section, provide all the entities/tables in a tabular format showing all the columns, keys, data types, etc. of each table. Must provide the Data Dictionary of each Entity set in your ERD.

[User]

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
user_id	int	11	Not Null	User Identification Number
user_name	varchar	255	Not Null	Users full name
user_fname	varchar	255	Not Null	Users first name
user_lname	varchar	255	Not Null	Users last name
user_mname	varchar	1	Not Null	Users middle name
user_email	varchar	255	Not Null	Users email address
user_pass	varchar	255	Not Null	Users password created
user_type	varchar	255	Not Null	User type
user_image	varchar	255	Not Null	User image
user_code	int	10	Not Null	User code

[Stocks Inventory]

ATTRIBUTE	DATA TYPE	FIELD SIZE	KEY TYPE	DESCRIPTION
item_id	int	255	Not Null	Item identification number
item_category	varchar	255	Not Null	Item category
item_name	varchar	255	Not Null	Item name
item_size	varchar	100	Not Null	Item size
Item_desc	varchar	255	Not Null	Item details
Item_qty	int	11	Not Null	Item quantity

CHAPTER III

RESULTS AND DISCUSSION

A. System Design

This section shows and describes the system following:

User Interface Design

Upon entering the system you will use the username and password created.

The screenshot displays the 'Add User' form within the 'Inventory System' interface. The form is titled 'Add User' and contains several input fields: 'First Name' (with a placeholder 'Enter Firstname'), 'Last Name' (with a placeholder 'Please fill out this field'), 'Middle Name' (with a placeholder 'Enter Middle Name'), 'Username' (with a placeholder 'Enter Username'), 'Email' (with a placeholder 'Enter Email Address'), 'Password' (with a placeholder 'Enter Password'), and 'Select User Type' (a dropdown menu with 'Admin' selected). There is also an 'Upload Image' section with a 'Choose File' button and 'No file chosen' text. At the bottom of the form are 'Save' and 'Cancel' buttons. A modal dialog box titled 'Confirm Register User' is overlaid on the form, featuring a question mark icon and 'Confirm' and 'Cancel' buttons. The background shows a sidebar with 'Main' menu items: 'Dashboard', 'Inventory', 'Report', and 'User' (selected). The main content area shows a 'Stock Info' table with columns 'NO.' and '+', and a table with columns 'EMAIL', 'USER TYPE', and 'ACTION'.

Figure 12. Sign Up

Log in after you have put the username and password you could press the login to enter the dashboard.

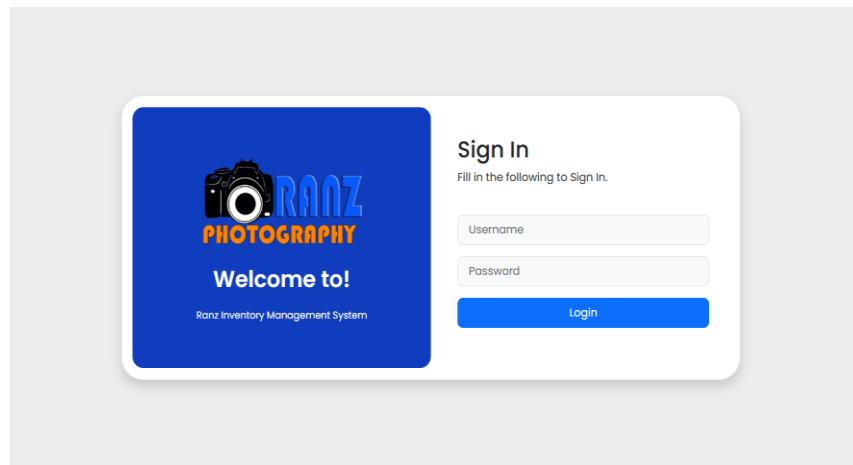


Figure 13. Log-in

When entering the dashboard, you will see "Welcome! Admin" including your name in the bottom. You can also see the total of stocks you have in the inventory, including the names and its total number of items.

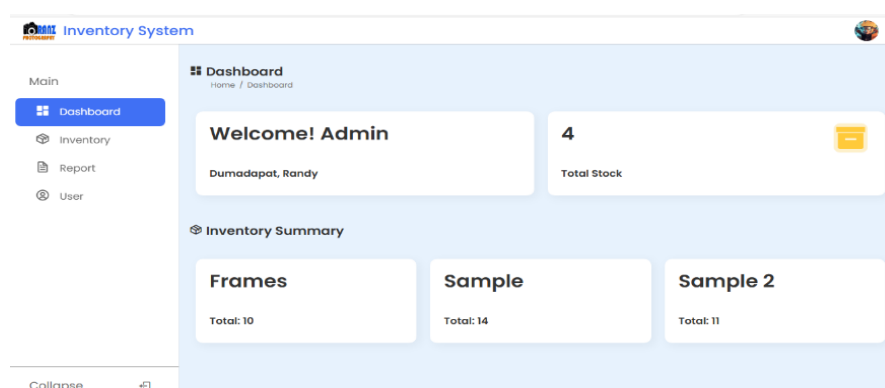


Figure 14. Dashboard

The update functionality, it can edit the details of the stocks for the possible misinformation data.

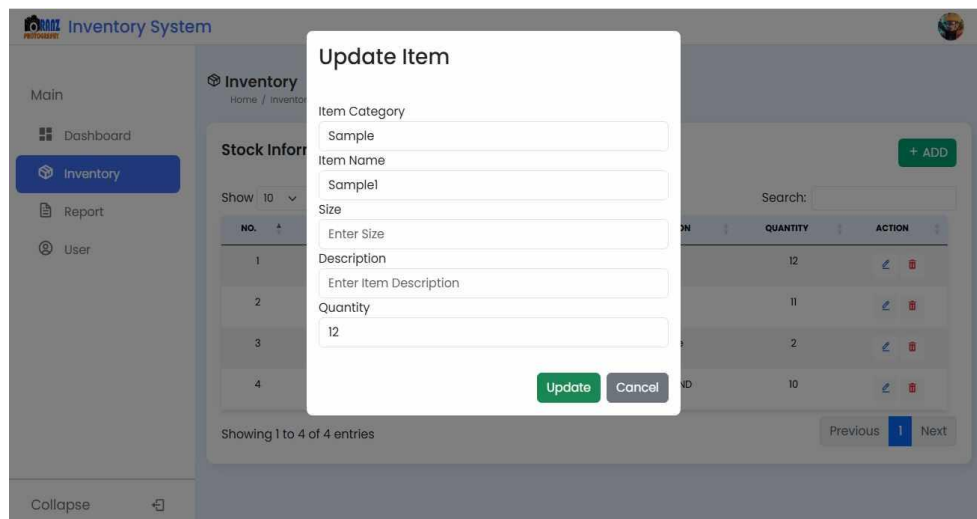


Figure 15. Update Item

The delete button will remove stocks from inventory, when the items are out of stocks.

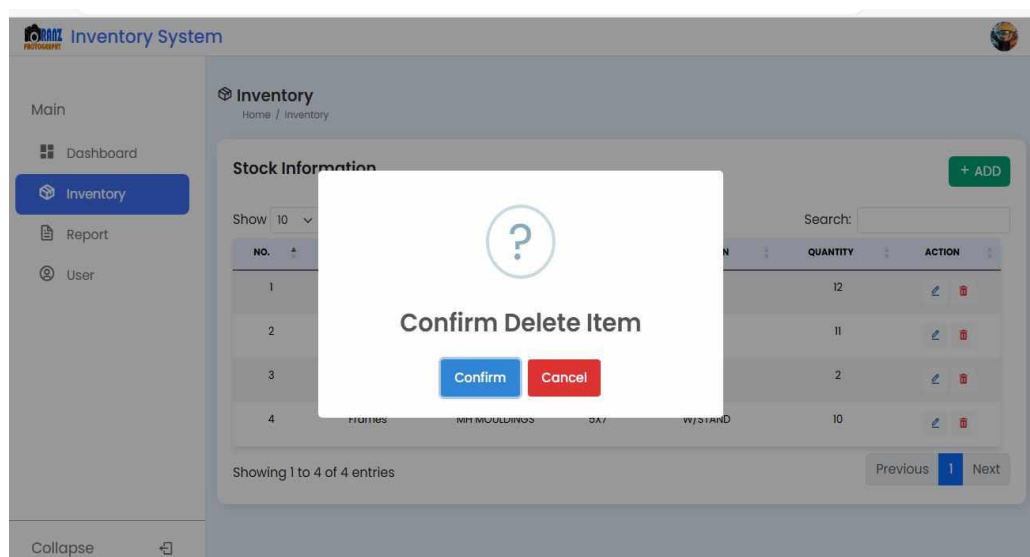


Figure 16. Delete Item

The add button will add stocks in the inventory, when items are available and purchased.

Add Item

Item Category
Enter Category

Item Name
Enter Item Name

Size
Enter Size

Description
Enter Item Description

Quantity
Enter item quantity

Save Cancel

Figure 17. Add Item

Inventory content are the overview list of stocks in the whole inventory process.

Inventory

Home / Inventory

Stock Information

Show 10 entries

Search:

+ ADD

NO.	CATEGORY	ITEM NAME	SIZE	DESCRIPTION	QUANTITY	ACTION
1	Sample	Sample1			12	Edit Delete
2	Sample 2	Bond Paper	A4		11	Edit Delete
3	Sample	Mug	medium	White	2	Edit Delete
4	Frames	MH MOULDINGS	5X7	W/STAND	10	Edit Delete

Showing 1 to 4 of 4 entries

Previous 1 Next

Figure 18. Inventory Details

Report content are the overall inventory of stocks report

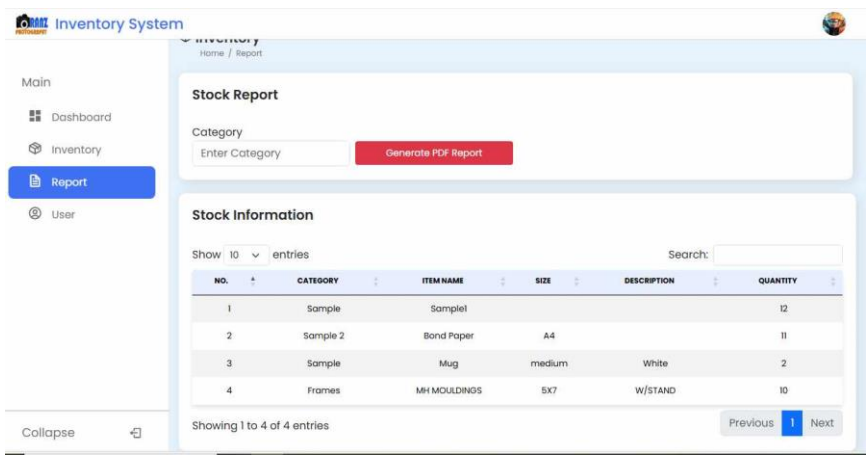


Figure 19. Inventory Report

User content is the system user details, including user name and password created.

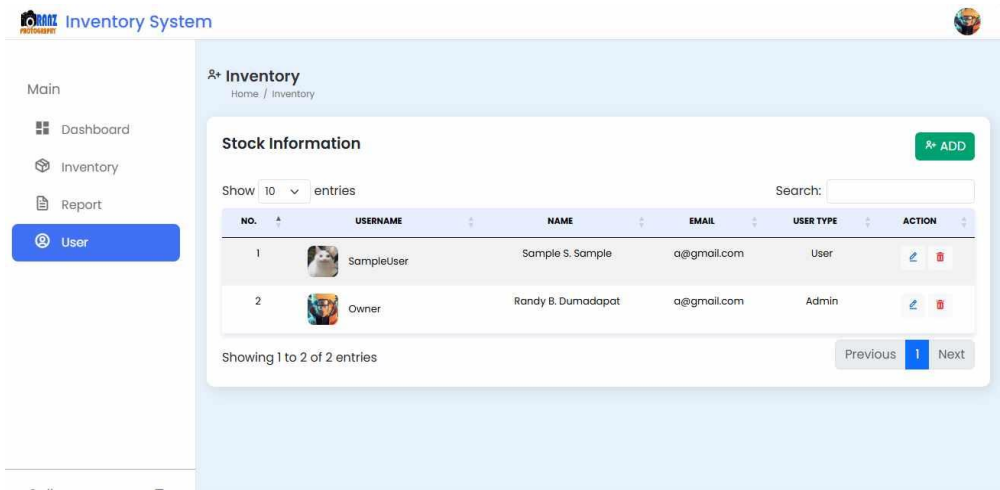


Figure 20. User Details

User profile content shows the user information and other details, including its photo.

 Inventory System

Main

Dashboard

Inventory

Report

User



DUMADAPAT, RANDY

User Information

First Name

Randy

Middle Name

B

Email

a@gmail.com

Upload Image

Choose File

No file chosen

Last Name

Dumadapat

Username

Owner

Password

Update

Figure 21. User Profile

CHAPTER IV

CONCLUSION AND RECOMMENDATION

A. Conclusion

This study has given the organization to eliminate their manual of writing down the inventory, including the hassle of its cause. This also highlights the manage time of its usage due to the overseeing inventory stocks. Moreover in conducting this research the organization come up from eliminating its hindrances over the pass years of service. This has also been helpful to them in managing their inventory which they add and remove items from their stocks, this leads to monitoring its entire inventory.

B. Recommendation

This study is recommended for its future purpose upon giving the organization a helpful way to eliminate their problems. This could also bring up the organization in which it is growing and adapting to the new change of software and technologies, including the future of this organization and the system that the researchers implemented, this has broaden the wide views for individuals in having Ranz photography as a their organization.

H. References

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APPENDICES

H. Letter to the Organization

Provide a scanned copy of the **APPROVED** letter to the organization. A Letter is proof that the project sponsor has approval to conduct environmental scanning of their organization.

I. Environmental Scanning Documentation

Provide pictures as proof that the project team conducted environmental scanning to the organization.

J. Curriculum Vitae

Provide a curriculum vitae of the project team.